

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 10

For the tutorial in Week 11 on 4 April, let us discuss the following questions.

Question 1. (Ex 10.7.8, 10.7.13.)

- (a) Suppose that a connected planar graph has eight vertices, each of degree three. Into how many regions is the plane divided by a planar representation of this graph ?
- (b) Suppose that a connected planar simple graph with m edges and n vertices contains no simple circuits of length 4 or less. Show that $m \leq (5/3)n - (10/3)$ if $n \geq 4$.

Solution:

- (a) Euler's Formula states that $r = m - n + 2$, and here we have $m = 3 \cdot 8/2 = 12$ edges and $n = 8$ vertices. Hence there are $r = 6$ regions.
- (b) When there are no simple circuits of length 4 or less, each region will be bounded by at least 5 edges. Recall that the sum of degrees of all regions is equal to twice the number of edges. Therefore,

$$2m = \sum_R \deg(R) \geq 5r,$$

which implies that $r \leq 2m/5$. By Euler's Formula, $m - n + 2 = r \leq 2m/5$, hence

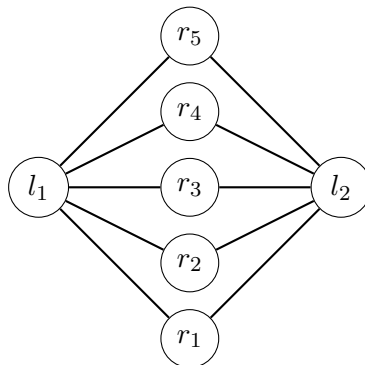
$$m \leq (5/3)n - (10/3).$$

Question 2.

- (a) Prove that $K_{2,5}$ is planar by drawing the graph without edges that cross each other. What about $K_{2,n}$?
- (b) Prove that there are no 6-regular simple graphs that are planar. (Hint: How many edges does a 6-regular graph have?)

Solution:

- (a) All $K_{2,n}$ are planar. Here is a drawing of $K_{2,5}$.



- (b) A 6-regular graph with n vertices has $3n$ edges by the handshaking lemma. But a planar graph has at most $3n - 6$ edges by a corollary of Euler's formula (Corollary 1 Section 10.7), which is a smaller number.

Question 3. Show that if G is a simple graph with at least 11 vertices, then either G or $\bar{G}$, the complement of G , is nonplanar.

Solution: Suppose both G and $\bar{G}$ are planar. Then from Corollary of Euler's formula (Corollary 1 Section 10.7), $m \leq 3n - 6$. For a graph G with m edges and n vertices, its complement $\bar{G}$ has $n(n-1)/2 - m$ edges. Hence, we have $m \leq 3n - 6$ and

$$\frac{n(n-1)}{2} - m \leq 3n - 6.$$

Adding both inequalities up, we arrive at

$$\frac{n(n-1)}{2} \leq 6n - 12,$$

which implies $n^2 - 13n + 24 \leq 0$. Solving this inequality, we see that n must satisfy $2.23 < n < 10.77$. Hence $n \leq 10$ which contradicts G having at least 11 vertices.

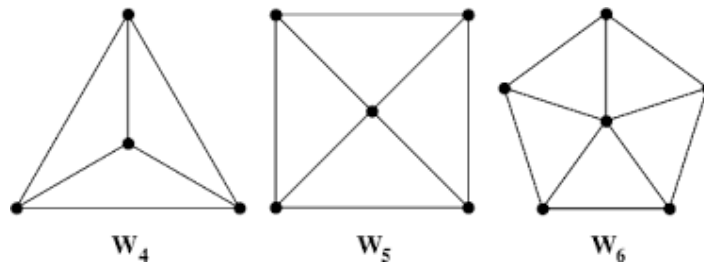
Question 4.

- (a) Find a planar 3-regular graph.
- (b) Is there a planar graph with 10 vertices and 25 edges?

Solution:

- (a) K_4 .
- (b) No. By Corollary of Euler's formula (Corollary 1 Section 10.7), the number of edges m satisfies the inequality $m \leq 3n - 6$. Hence such a graph can have at most $3 \cdot 10 - 6 = 24$ edges.

Question 5. Determine the chromatic numbers of the following three graphs (called *wheel graphs*).



Solution: The wheel graphs are cycles together with a central vertex that is connected to all other vertices, and hence needs an additional color for the central vertex. The chromatic numbers of C_3, C_4 , and C_5 are 3, 2, 3, respectively. So the chromatic numbers of W_4, W_5 , and W_6 are 4, 3, 4, respectively.

Question 6.

- (a) Under what condition on n is there a graph with n vertices where all vertices have degree 1?
- (b) True or false: a planar graph that has no subgraph that is an odd cycle can be 2-colored.

Solution:

- (a) Such a graph exists if and only if $n \geq 2$ is even. By the handshaking lemma, n cannot be odd. Also, when n is even we can pair vertices and connect the pairs by an edge each.
- (b) True. A graph that has no odd cycle is bipartite and can be colored with two colors, whether planar or not.

Question 7.

- (a) How many edges does one have to add to the wheel graph W_5 (see above) so that the resulting (simple) graph is non-planar?
- (b) If we add three new edges to W_6 (so that the result is a simple graph), can the result be planar?

Solution:

- (a) Since the graph has 5 vertices, the only way to make it non-planar, is to add edges until the graph becomes K_5 . There are 2 edges in K_5 that are not in W_5 , hence the answer is 2.
- (b) No. If the result would be planar, then this would be a planar graph with 6 vertices and 13 edges. According to Corollary of Euler's formula (Corollary 1 Section 10.7), the number of edges in a planar graph with 6 vertices can be at most $3 \cdot 6 - 6 = 12$.

Question 8.

- (a) Is there a connected planar graph with n vertices and $2n$ regions?
- (b) Is there a connected planar graph with n vertices and n regions?

Solution:

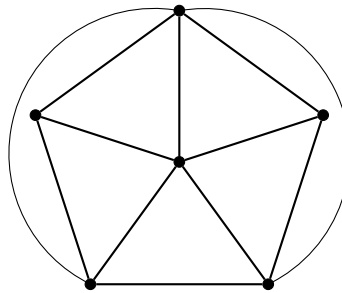
- (a) No. According to the corollary of Euler's formula ($m \leq 3n - 6$), such a graph can have at most $3n - 6$ edges. Applying this to Euler's formula ($r = m - n + 2$), we see that there are at most $3n - 6 - n + 2 = 2n - 4$ regions.
- (b) Yes. The wheel graph W_n (a cycle C_{n-1} together with an extra vertex connected to all other vertices) has n regions.

Question 9.

- (a) Let $n \geq 3$. Is there a connected, planar, bipartite graph with n regions and n vertices?
 (b) Is there a connected planar graph with 6 vertices and 8 regions?

Solution:

- (a) No. According to a Corollary of Euler's formula (Corollary 3 Section 10.7), such a graph has at most $2n - 4$ edges. Applying this to Euler's formula ($r = m - n + 2$), there are at most $r = 2n - 4 - n + 2 = n - 2$ regions.
 (b) Yes. Take the wheel W_6 , and add two "ears" on the outside. The ears must be distinct, new edges.

**Answer.**

- 1(a) 6.
 2(a) planar.
 5) 4, 3, 4.
 6(b) True.
 7) 2; No.
 8) No; Yes.
 8) No; Yes.